

an algorithmic solution. Comparing these solutions will provide a vivid illustration of what an algorithm is and how it's different from other approaches to problem-solving. Additionally, the outfielder problem will help us visualize a field that is occasionally abstract—you probably have some experience throwing and catching something, and this experience can help you understand the theory behind your practice.

Before we can really understand how a human knows exactly where a ball will land, it will help to understand how a machine does it. We'll start by looking at an analytic solution to the outfielder problem. This solution is mathematically precise and easy for computers to execute instantaneously, and some version of it is usually taught in introductory physics classes. It would enable a sufficiently agile robot to play outfield for a baseball team.

However, humans can't easily run analytic equations in their heads, and certainly not as quickly as computers can. A solution that's better suited to human brains is an algorithmic solution, which we'll use to explore what an algorithm is and what its strengths are compared to other problem-solving solutions. Moreover, the algorithmic solution will show us that algorithms are natural to human thought processes and don't need to be intimidating. The outfielder problem is meant to introduce a new way to solve problems: the algorithmic approach.

## The Analytic Approach

To solve this problem analytically, we have to go back a few centuries to an early model of motion.

### *The Galilean Model*

The equations most commonly used to model a ball's movement date back to Galileo, who centuries ago formulated polynomials that capture acceleration, speed, and distance. If we ignore wind and air resistance and assume the ball starts at ground level, Galileo's model says that the horizontal position of a thrown ball at time  $t$  will be given by the formula

$$x = v_1 t$$